Healthcare Appointment No-Show Prediction

📌 Objective

This project aims to analyse patient behaviour in healthcare scheduling to identify key factors driving missed appointments and build a predictive model that helps clinics improve operational efficiency through data-driven strategies.

🛠 Tools Used

- **Python:** Pandas, Scikit-learn
- **Machine Learning:** Decision Tree Classifier
- **Data Visualisation:** Power BI

📂 Data Preprocessing & Visualization

Dataset Overview The dataset contains patient-level information, including demographics, appointment details, and SMS reminder logs. Each record represents an individual appointment, with a target variable indicating the patient's attendance status.

Key Variables Age, Gender, SMS_received, WaitingDays, Appointment Weekday, and Attendance Status (target variable).

⚙️ Methodology

1. Data Preprocessing

- Handled missing values to ensure data integrity.
- Engineered new features, notably WaitingDays, to capture the time between scheduling and the actual appointment.
- Converted categorical variables into numerical formats for model compatibility.

2. Exploratory Data Analysis & Dashboarding An interactive Power BI dashboard was developed to analyze attendance patterns across:

- Patient age demographics
- Appointment weekdays
- The impact of SMS reminders
- The correlation between waiting time and no-show rates

3. Predictive Modeling A Decision Tree Classifier was developed to predict whether a patient will miss an appointment based on the preprocessed features.

📊 Key Findings

- The overall no-show rate stands at approximately 20%.
- Longer waiting days strongly increase the likelihood of missed appointments.
- Sending SMS reminders effectively reduces the no-show rate.

- Younger patient demographics exhibit higher no-show rates compared to other age groups.

🚀 Conclusion

This project demonstrates how data analytics and machine learning can effectively identify the drivers behind missed healthcare appointments. By leveraging this predictive model and actionable dashboard insights, healthcare providers can proactively optimize scheduling systems and implement targeted reminder campaigns to reduce no-shows significantly.